Fast and Interpretable Dynamics for Fisher Markets via Block-Coordinate Updates

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MAIN CONTRIBUTIONS

- Two stochastic algorithms for computing large-scale market equilibria;
- Strong theoretical convergence guarantees and details for efficient implementation;
- A tâtonnement-style interpretation for projected-gradient-type methods in ME computation;
- Extensive numerical experiments demonstrating the numerical efficiency of the algorithms.

Block Coordinate Descent for EG (**BCDEG**)

$$\begin{aligned}\max_x \quad & \sum_i B_i \log u_i \\ \text{s.t.} \quad & u_i \le \langle v_i, x_i\rangle \quad \forall\, i \\ & \sum_i x_{ij} \le 1 \quad \forall\, j \\ & x \ge 0.\end{aligned} \quad \rightarrow \quad \begin{aligned}\min_x \quad & F(x) := f(x) + \sum_j \psi_j(x_{\cdot j}) \\ & f(x) = -\sum_i B_i \log \langle v_i, x_i\rangle \\ & \psi_j(x_{\cdot j}) = \begin{cases} 0 & \text{if } x_{\cdot j} \in \Delta_n \\ +\infty & \text{otherwise}\end{cases} \quad \forall\, j\end{aligned}$$

- At each iteration, randomly choose $j \in [m]$ and only update coordinate j via

$$x_{\cdot j}^+ = T_j(x) = \arg\min_{y_{\cdot j}} \langle \nabla_{\cdot j} f(x), y_{\cdot j} - x_{\cdot j}\rangle + \frac{1}{\eta_j}\|y_{\cdot j} - x_{\cdot j}\|^2 + \psi_j(y_{\cdot j})$$

or equivalently

$$T_j(x) = \text{Proj}_{\Delta_n}\left(x_{\cdot j} - \eta_{j'} \nabla_{\cdot j} f(x)\right)$$

Lemma (Coordinate-wise Lipschitz Continuity).
For any $j \in [m]$, let

$$L_j = \max_{i\in[n]} \frac{B_i v_{ij}^2}{\underline{u}_i^2}. \qquad (*)$$

Then, for all $x, y \in \mathcal{X}$ such that x, y differ only in the jth block, we have

$$\tilde{f}(y) \le \tilde{f}(x) + \langle \nabla_{\cdot j}\tilde{f}(x), y - x\rangle + \frac{L_j}{2}\|y - x\|^2.$$

Theorem (Linear Convergence of BCDEG).
Given an initial iterate x^0 and stepsizes $\eta_j^0 = 1/L_j, \forall\, j$ satisfying (*), let x^k be the random iterates generated by BCDEG. Then,

$$\mathbf{E}\left[F(x^{k+1})\right] - F^* \le (1-\rho)^k \left(F(x^0) - F^*\right),$$

where $\rho = \min\left\{\frac{\mu}{mL_{\max}\theta^2(A,C)}, \frac{1}{m}\right\}$ and $L_{\max} = \max_j L_j$.

Fisher Market, Market Equilibrium

In a Fisher market:

- There is a set of m divisible items (each denoted as j) with unit supplies;
- There is a set of n buyers (each denoted as i) with budgets $B_i > 0$.

A market equilibrium (ME) is an allocation-price pair satisfying

- Buyer optimility;
- Market clearance.

Block Coordinate Proportional Response (**BCPR**)

$$\begin{aligned}\max_b \quad & \sum_{i,j:v_{ij}>0} b_{ij} \log\left(\frac{v_{ij}}{p_j(b)}\right) \\ \text{s.t.} \quad & \sum_j b_{ij} = B_i \quad \forall\, i \\ & p_j(b) = \sum_i b_{ij} \quad \forall\, j \\ & b \ge 0.\end{aligned} \quad \rightarrow \quad \begin{aligned}\min_b \quad & \Phi(b) := \varphi(b) + \sum_i r_i(b_i) \\ & \varphi(b) = -\sum_{i,j:v_{ij}>0} b_{ij} \log\left(\frac{v_{ij}}{p_j(b)}\right) \\ & r_i(b_i) = \begin{cases} 0 & \text{if } \sum_j b_{ij} = B_i, b_i \ge 0 \\ +\infty & \text{otherwise}\end{cases} \quad \forall\, i\end{aligned}$$

- At each iteration, randomly choose $i \in [n]$ and only update coordinate i via

$$b_i^+ = T_i(b) := \arg\min_{a_i} \langle \nabla_i \varphi(b), a_i - b_i\rangle + \frac{1}{\alpha_i} D_{\text{KL}}(a_i, b_i) + r_i(a_i)$$

or equivalently

$$b_{ij}^+ = \frac{1}{Z_i} b_{ij}\left(\frac{v_{ij}}{p_j}\right)^{\alpha_i}, \text{ where } Z_i \text{ is a normalization constant s.t.} \sum_j b_{ij}^+ = B_i.$$

Theorem ($O(1/k)$ Convergence of BCPR).
Let b^k be random iterates generated by BCPR with $\alpha_i^k = 1/L_i, \forall\, i$ for all k, then

$$\mathbf{E}\left[\Phi(b^k)\right] - \Phi^* \le \frac{n}{n+k}\left(\Phi^* - \Phi(b^0) + D(b^*, b^0)\right),$$

where Φ^* is the optimal objective value.

- We also give a Mirror Descent with Line Search (MDLS) algorithm, leveraging recent advances in relatively smooth convex optimization.

Theorem ($O(1/k)$ Convergence of PRLS).
Let b^k be iterates generated by PRLS starting from any initial feasible solution b^0, then we have

$$\Phi(b^k) - \Phi^* \le \frac{1}{\rho^-} \cdot \frac{D(b^*, b^0)}{k},$$

where ρ^- is the shrinking factor of stepsize.

Numerical Experiments

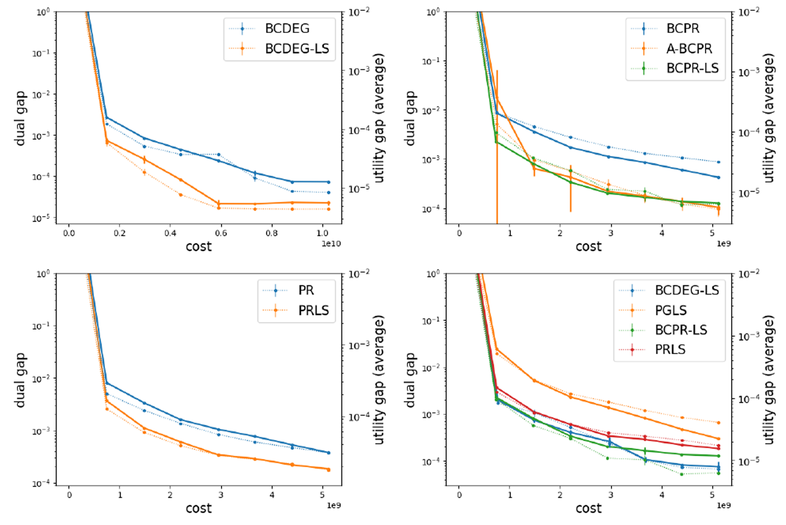

Figure 1 Experiment results on simulated data

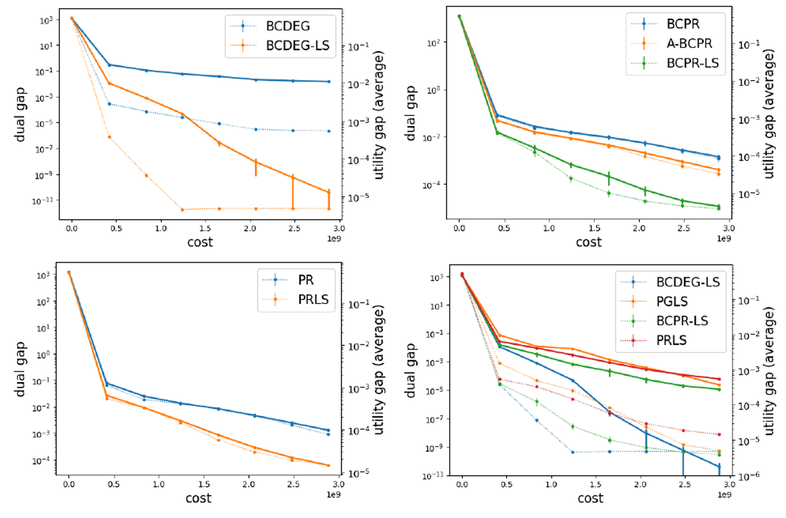

Figure 2 Experiment results on real data

Economic Interpretation of Projected Gradient Steps

- Projected gradient step for buyer i and item j:

$$y = x_{\cdot j} - \eta_j g \text{ and } x_{\cdot j}^+ \leftarrow \text{Proj}_{\Delta^n}(y).$$

- The key step in projection computation: find the unique multiplier t such that

$$\sum_i (y_i - t)^+ = 1,$$

which gives the solution $x_j^+ = (y - t\cdot\mathbf{1})^+$.

- Replacing t with $\eta_j p_j$ for a "price" p_j:

$$x_{ij}^+ = D_i(p_j) = (x_{ij} - \eta_j g_i - \eta_j p_j)^+ \text{ and } \sum_i D_i(p_j) = 1.$$

  - D_i: buyer i's linear demand finction given prior allocation x_i
  - p_j: market-clearing price for item j

Seller-side "dynamic pricing"

- Randomly sample an item j
- Each buyer i forms a demand function of item j:

$$D_i(p_j) \leftarrow (x_{ij} - \eta_j g - \eta_j p_j)^+$$

- Seller sets the unique (market-clearing) price p_j such that

$$\sum_i D_i(p_j) = 1$$

- Each buyer i chooses her new allocation of item j via her demand

$$x_{ij}^+ = D_i(p_j)$$

Let $\beta_i = B_i/u_i$ be the current utility price for buyer i. Then:

- $x_{ij}^+ = x_{ij}$ (stays the same) if $p_j = \beta_i v_{ij}$;
- $x_{ij}^+ > x_{ij}$ (increases) if $p_j < \beta_i v_{ij}$;
- $x_{ij}^+ < x_{ij}$ (decreases) if $p_j > \beta_i v_{ij}$.

- Buyers slowly adapt to the changing environment, while the prices are set to balance supply and demand, similar to a **tâtonnement** process.